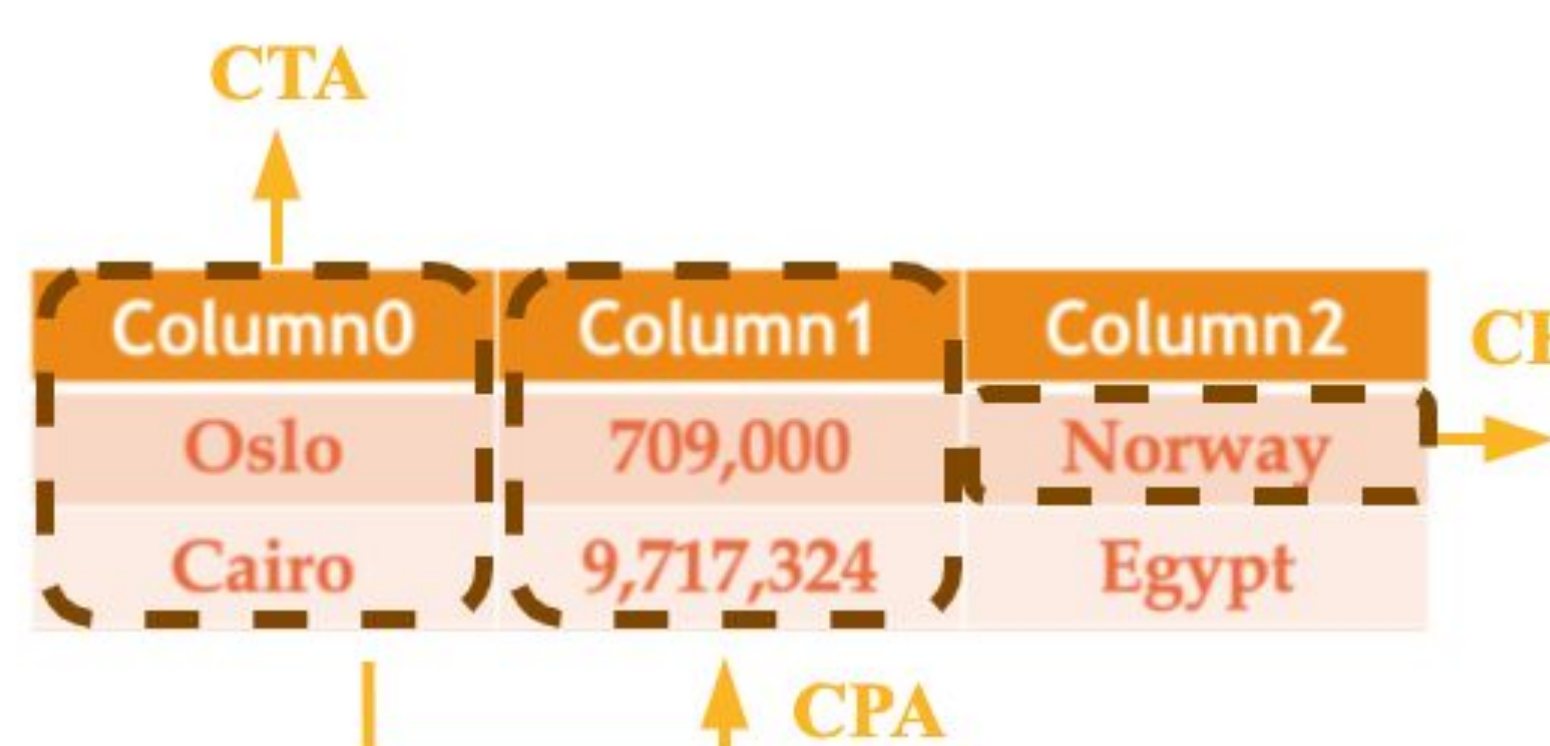


TorchicTab: Semantic Table Annotation with Wikidata & Language Models

Semtab Tasks

CTA - Assign a semantic **type** to a columnCPA - Assign a **property** to the relationship between two columnsCEA - Match a cell to a Wikidata **entity**

TorchicTab Systems

Is RDF graph querying required?

YES

Tasks

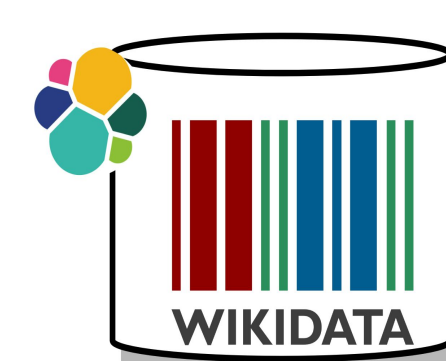
NO

Heuristic Method

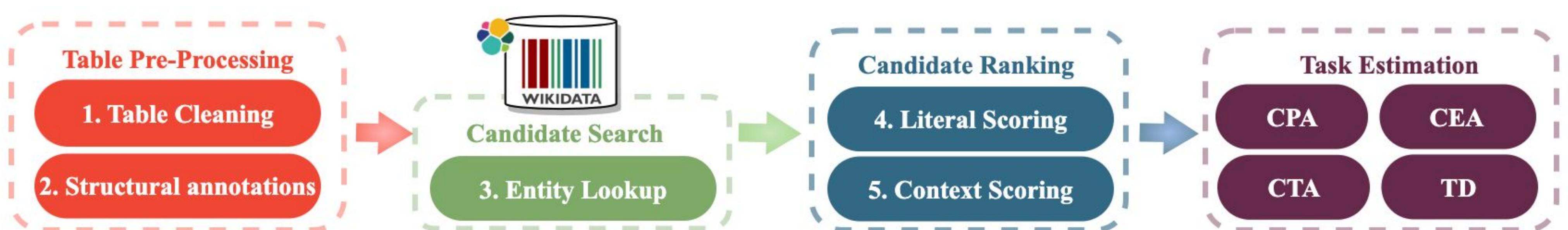
Wikidata

Classification

Language Models



TorchicTab - Heuristic



1) Table Cleaning

- remove noisy data, e.g., HTML tags
- fix incorrect encodings

2) Structural annotation

- classify cells as named entities (NE) or literal values (L)
- assign NE-column or L-column by majority voting

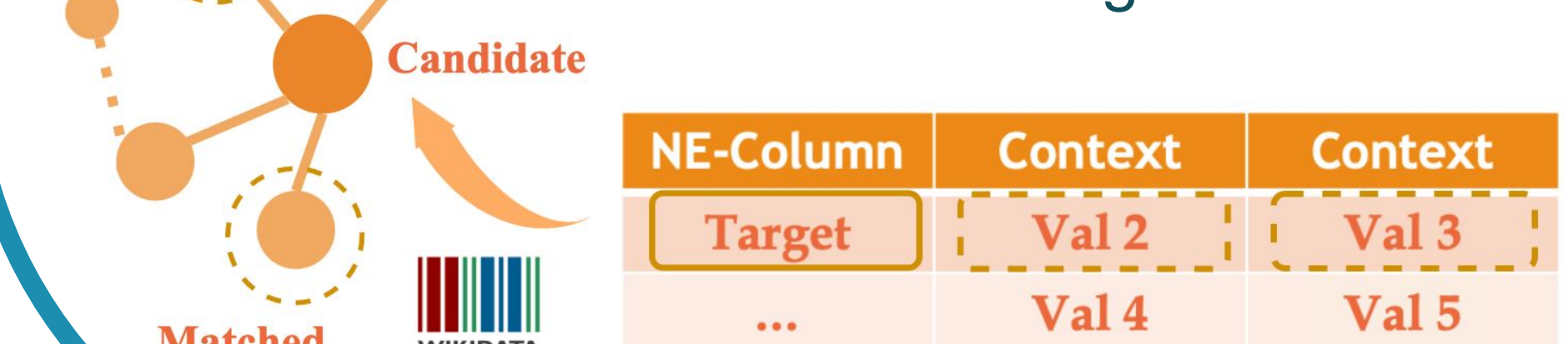
3) Entity Lookup BM25 & Levenshtein ratio

Retrieve candidates using cells inside NE-columns

- Complete Cell Search
- Fuzzy Search
- Cell Token Search (individual)
- Cell Token Search (combination)

4) & 5) Candidate Ranking (similarity)

- Between target cell & candidates
- Between context cells & candidate's neighborhood



6) CPA

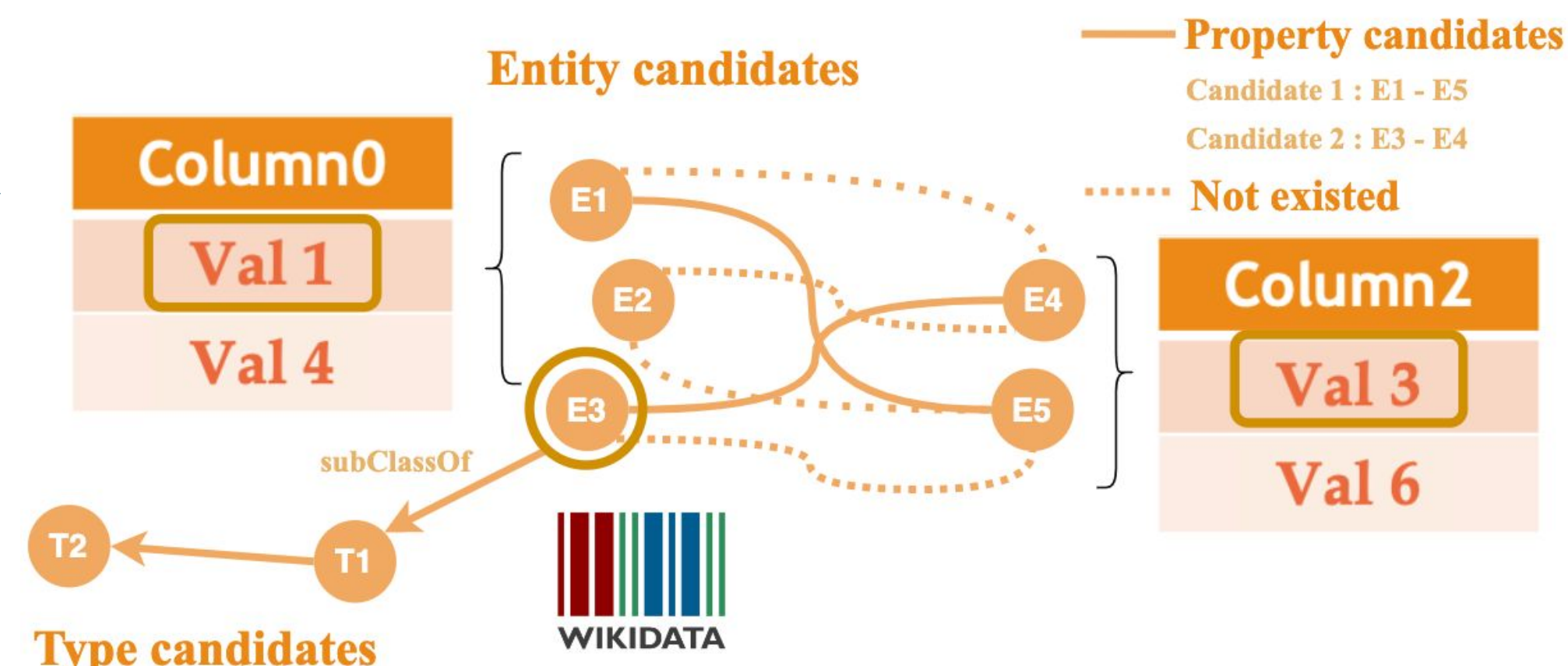
- iterate all the combinations of the entity candidates from two columns to retrieve possible property candidates
- rank property candidates
 - previous entity candidates' scores
 - property occurrence

7) CEA

- rank entity candidates: previous score & CPA result

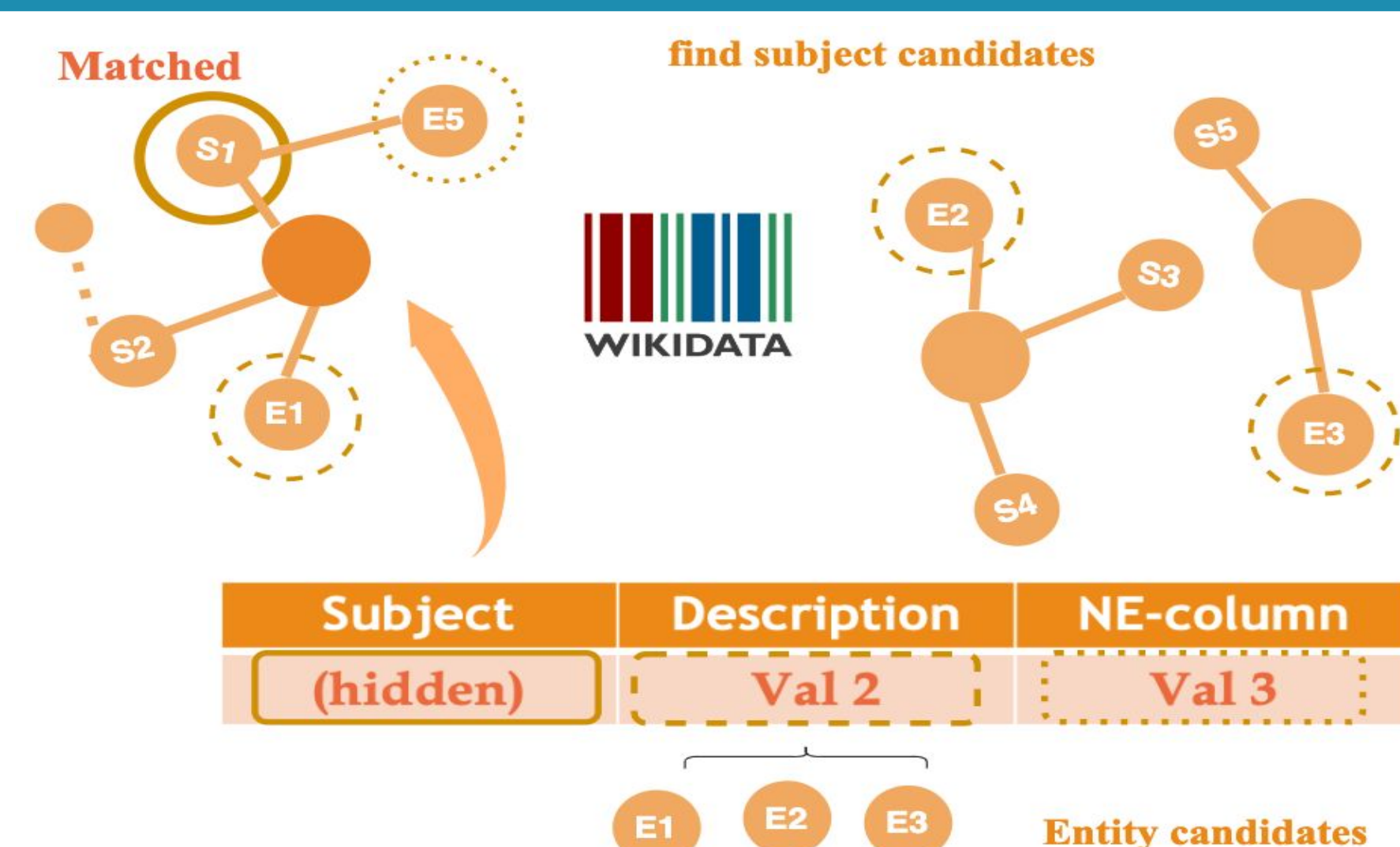
8) CTA

- iterate 'instance of' & 'subclass of' with chosen entities to retrieve type candidates
- rank type candidates: node distance & type occurrence



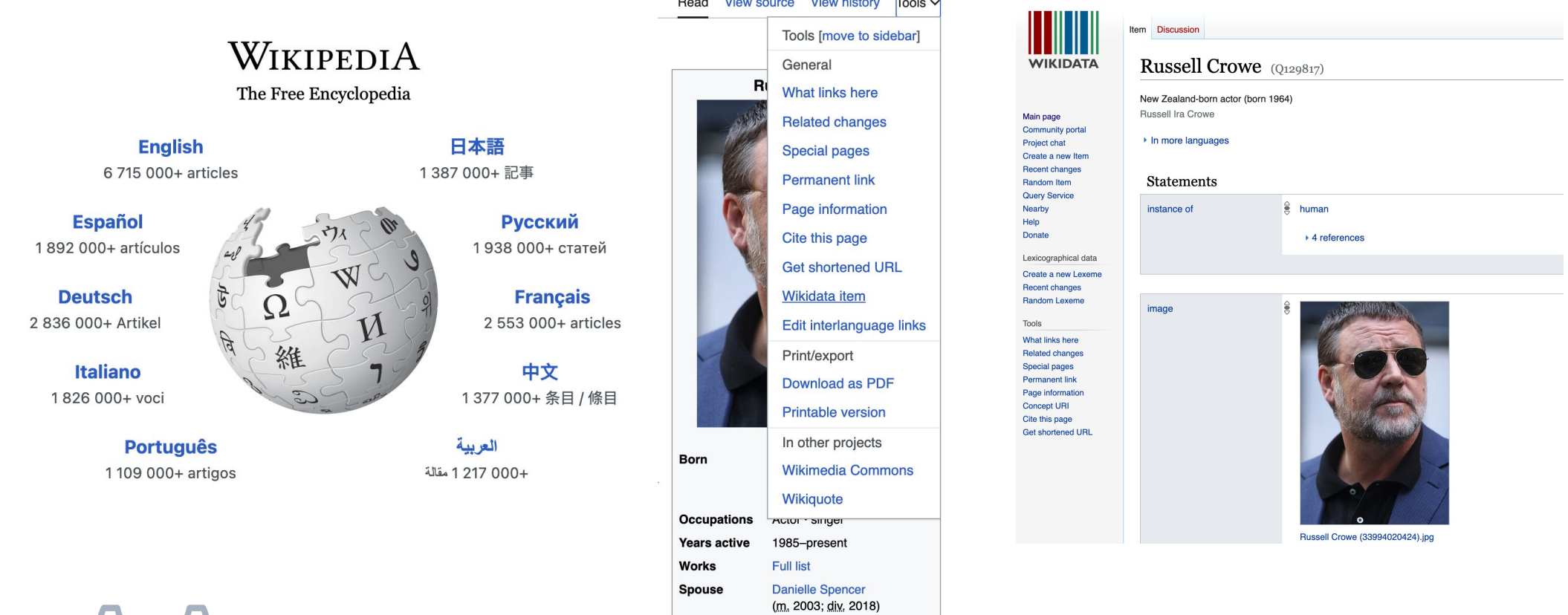
9) TD (see the figure in the upper right corner)

- iterate all the 'superclass of' the description's candidates to retrieve their subclasses as topic candidates
- rank topic candidates: occurrence & context score



10) CQA

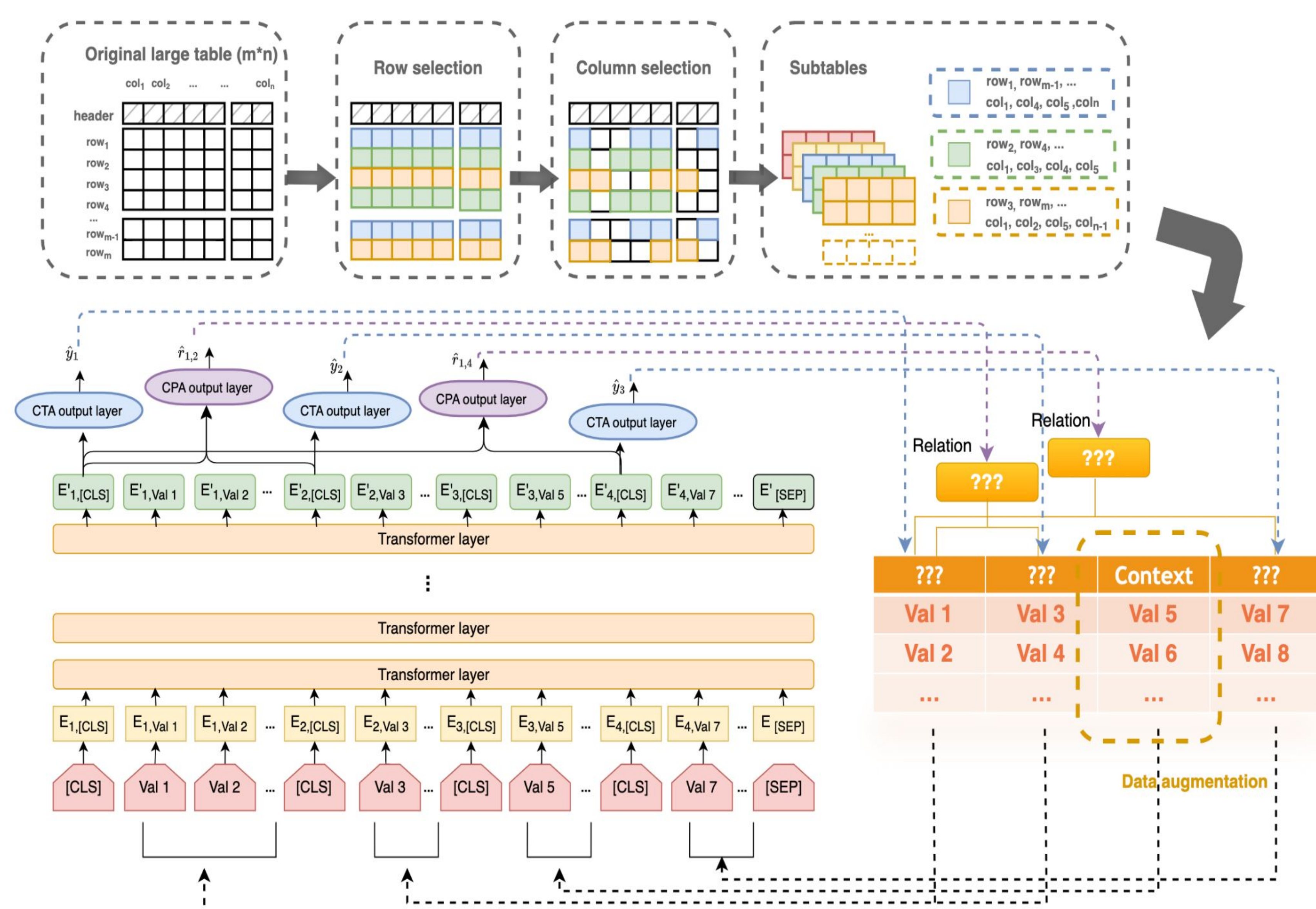
- apply redirect function inside wikipedia "Tools - wikidata item"
- retrieve qualifier candidates with SPARQL query
- rank the qualifier candidates: occurrence



Round1 & CQA - Results

Benchmark	Target	Task	F1	Precision
WikidataTables	Wikidata	CEA	0.830	0.839
		CTA	0.817	0.749
		CPA	0.934	0.934
CQA	Wikidata	CQA		0.822

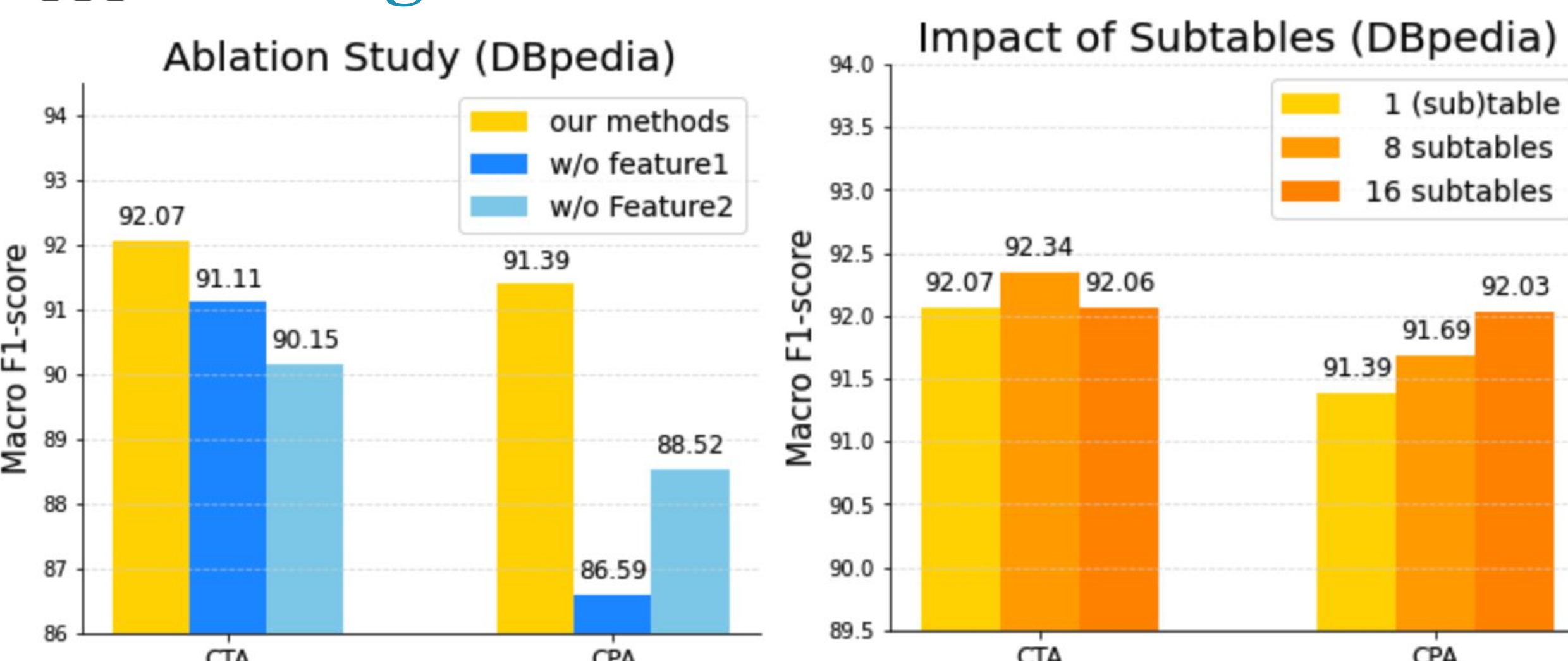
TorchicTab - Classification



Features

- Subtable sampling - split a large table into small subtables
- Data augmentation - enrich context with more columns

Findings



Round2 - SOTAB - Results

Benchmark	Target	Task	F1-Score	Precision
SOTAB	Schema.org	CTA	0.8966	0.8980
		CPA	0.8711	0.8800
SOTAB	DBpedia	CTA	0.8972	0.9093
		CPA	0.9049	0.9068